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Studierichting: Informatica
Oefeningenreeks 5D nr. 5.1

$$y^2 = 2x - 1 \text{ en } y = 2x - 3$$

$$x_1 = \frac{y^2+1}{2} \text{ en } x_2 = \frac{y+3}{2}$$

$$\Rightarrow S = \int_{-1}^{2} \left(\frac{y+3}{2} - \frac{y^2+1}{2} \right) dy$$

$$= \int_{-1}^{2} \frac{1}{2} \left(-y^2 + y + 2 \right) dy$$

$$= \left[-\frac{y^3}{6} + \frac{y^2}{4} + y \right]_{-1}^{2}$$

$$= -\frac{4}{3} + 1 + 2 - \left(\frac{1}{6} + \frac{1}{4} - 1 \right)$$

$$= \frac{27}{12}$$

$$= \frac{9}{4}$$

References

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